ARJUNA JEE 2027

Lecture -07 | Mathematics

TRIGONOMETRIC FUNCTIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Transformation Formula's**

B **Question Practice**

C

D





LAST CLASS REVISION

Each Ques^{n}.
⇓
• S.A.S. •

Answer the following Questions:

(1) If value at angle $(\alpha + \beta)$ & $(\alpha - \beta)$ is given then

value at angle 2α = $(\alpha+\beta+\alpha-\beta)$.

and value at angle 2β = $(\alpha+\beta)-(\alpha-\beta)$.

(2) While solving questions involving tan & cot one can start with

angles.

LAST CLASS REVISION

Answer the following Questions:

(3) If $\alpha + \beta = \frac{\pi}{4}$ or $\frac{5\pi}{4}$ or $\frac{9\pi}{4}$, or $\frac{-3\pi}{4}$ or $-\frac{7\pi}{4}$ ..... then

$(1 + \tan\alpha)(1 + \tan\beta) =$ 2 .

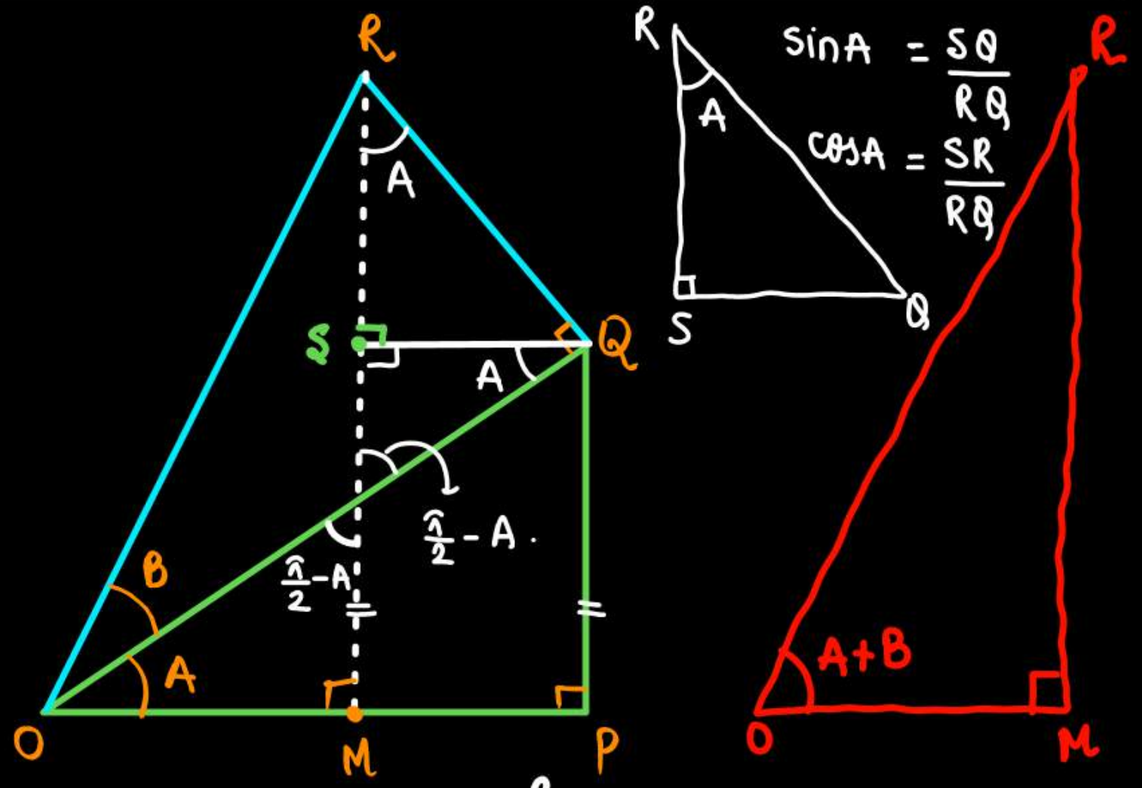
$1 = \tan\frac{\pi}{4} = \tan\frac{5\pi}{4} = \tan\frac{9\pi}{4} = \tan\left(\frac{-3\pi}{4}\right) = \ldots$

$\tan\left(\pi + \frac{\pi}{4}\right)$ $\quad$ $\tan\left(2\pi + \frac{\pi}{4}\right)$ $\quad$ $-\tan\frac{3\pi}{4}$

$\tan\frac{\pi}{4}$ $\quad$ $\tan\frac{\pi}{4}$ $\quad$ $-\tan\left(\pi - \frac{\pi}{4}\right)$

$-\left(-\tan\frac{\pi}{4}\right)$

PROOF OF COMPOUND ANGLE FORMULA



$$\sin A = \frac{SQ}{RQ}$$

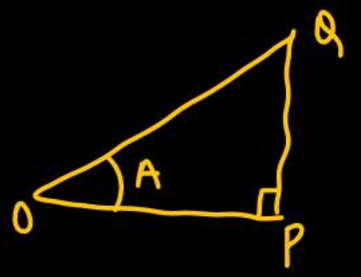
$$\cos A = \frac{SR}{RQ}$$

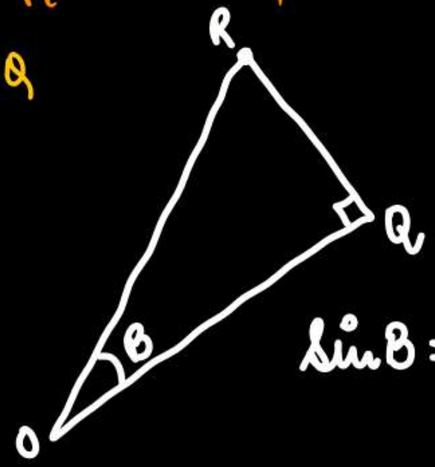
$$\sin(A+B) = \frac{RM}{OR} = \frac{RS}{OR} + \frac{SM}{OR}$$

PQ

$$= \frac{RS}{RQ} \times \frac{RQ}{OR} + \frac{PQ}{OQ} \times \frac{OQ}{OR}$$

$$= \cos A \cdot \sin B + \sin A \cdot \cos B$$





$$\cos B = \frac{OQ}{OR}$$

$$\sin B = \frac{RQ}{OR}$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\# \quad A \to \frac{\pi}{2} + A$$

$$\sin\left(\frac{\pi}{2} + \overbrace{A+B}^{\theta}\right) = \sin\left(\frac{\pi}{2}+A\right)\cos B + \cos\left(\frac{\pi}{2}+A\right)\sin B$$

$$\boxed{\cos(A+B) = \cos A \cos B - \sin A \sin B}$$

Question Practice

Question [JEE Mains-2019]

If $\cos(\alpha+\beta)=\frac{3}{5}, \sin(\alpha-\beta)=\frac{5}{13}$ and $0<\alpha, \beta<\frac{\pi}{4}$ then $\tan(2\alpha)$ is equal to

A $\frac{21}{16}$

B $\frac{33}{52}$

C $\frac{63}{16}$

D $\frac{63}{52}$

Done !!

Question [JEE Mains-2022]

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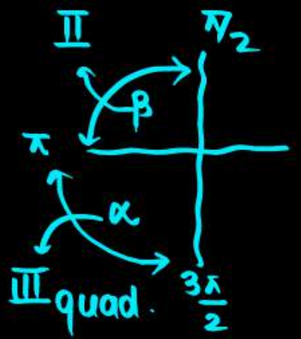
If $\cot\alpha = 1$ and $\sec\beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha+\beta)$ and the quadrant in which $\alpha+\beta$ lies, respectively are:

A ~~$-\frac{1}{7}$ and IVth quadrant~~

B ~~7 and Ist quadrant~~

C ~~-7 and IVth quadrant~~

D ~~$\frac{1}{7}$ and Ist quadrant~~



$\cot\alpha = 1$

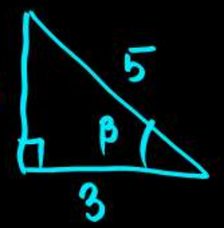
$\downarrow$

$\frac{1}{\tan\alpha} = 1 \rightarrow \tan\alpha = 1$

$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$ (1 over $\tan\alpha$, 1 over $\tan\beta$)

$= \frac{1 - \frac{4}{3}}{1 - 1\left(-\frac{4}{3}\right)} = \frac{-\frac{1}{3}}{\frac{7}{3}} = -\frac{1}{7}$

$\sec\beta = -\frac{5}{3} = \frac{H}{B}$



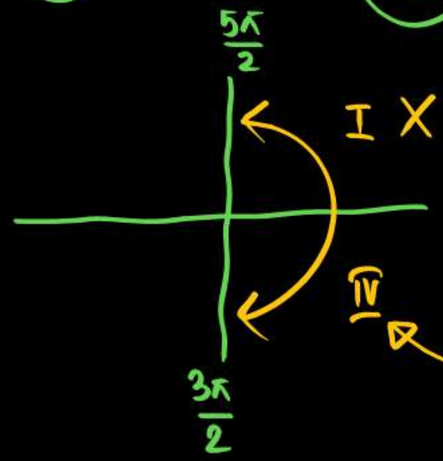
$\tan\beta = -\frac{4}{3}$

$\pi < \alpha < \frac{3\pi}{2}$

$\frac{\pi}{2} < \beta < \pi$

add

$\frac{3\pi}{2} < \alpha+\beta < \frac{5\pi}{2}$



$\tan(\alpha+\beta) = \ominus \frac{1}{7}$

Question

Show that : $\tan 3A - \tan 2A - \tan A = \tan A \tan 2A \tan 3A$

$3A = 2A + A$.

$\tan(3A) = \tan(2A + A)$

$\tan(3A) = \dfrac{\tan 2A + \tan A}{1 - \tan A \cdot \tan 2A}$

HHPP.

Question

If $A + B = 45°$, then prove that : $(\cot A - 1)(\cot B - 1) = 2$

$$(1 + \tan A)(1 + \tan B) = 2$$

$$\left(1 + \frac{1}{\cot A}\right)\left(1 + \frac{1}{\cot B}\right)$$

$$(\cot A + 1)(\cot B + 1) = 2\cot A \cot B$$

$$\cot A \cot B + \cot A + \cot B + 1 = 2\cot A \cot B$$

$$-1 + \cot A + \cot B - \cot A \cot B = -1 - 1$$

$$-(1 - \cot A) + \cot B(1 - \cot A) = -2$$

$$(1 - \cot A)(-1 + \cot B) = -2$$

$$+(\cot A - 1)(\cot B - 1) = +2$$

Question

If A + B = 45°, then prove that : (cot A – 1) (cot B – 1) = 2

M-2

$A + B = 45^\circ$

$\cot(A+B) = \cot 45^\circ$

$$\frac{\cot A \cot B - 1}{\cot A + \cot B} = 1$$

$\alpha\beta - 1 = \alpha + \beta$

$-1 = \alpha + \beta - \alpha\beta$

$-1 - 1 = -1 + \alpha + \beta - \alpha\beta$

$-2 = -(1-\alpha) + \beta(1-\alpha)$

$-2 = (1-\alpha)(-1+\beta)$

$+2 = +(\alpha-1)(\beta-1)$

IMPORTANT RESULT

RESULT-01:

If $A + B = \frac{\pi}{4}$, then $(1 + \tan A) \cdot (1 + \tan B) = 2$

RESULT-02:

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - (\tan A \tan B + \tan A \tan C + \tan B \tan C)}$$

RESULT-03:

$$\tan(\theta_1 + \theta_2 + \theta_3 + \ldots\ldots \theta_n) = \left(\frac{S_1 - S_3 + S_5 - S_7 + \ldots}{1 - S_2 + S_4 - S_6 + \ldots}\right)$$

$$\tan(\overbrace{A+B}^{\theta_1}+\overbrace{C}^{\theta_2}) = \frac{\tan(A+B)+\tan C}{1-\tan(A+B)\tan C}$$

$$= \frac{\left(\frac{\tan A+\tan B}{1-\tan A\tan B}\right)+\tan C}{1-\left(\frac{\tan A+\tan B}{1-\tan A\tan B}\right)\tan C} = \frac{\overbrace{\tan A+\tan B+\tan C}^{S_1} - \overbrace{\tan A\tan B\tan C}^{S_3}}{1-\underbrace{(\tan A\tan B+\tan A\tan C+\tan B\tan C)}_{S_2}}$$

$$\tan A\ \tan B\ \tan C. = \frac{S_1 - S_3}{1-S_2}$$

$$S_1 = \tan A+\tan B+\tan C.$$

$$S_2 = \tan A\tan B+\tan B\tan C+\tan A\tan C.$$

$$S_3 = \tan A\tan B\tan C.$$

$$\tan(A+B+C+D) = \frac{S_1 - S_3}{1 - S_2 + S_4}$$

$\alpha\beta\gamma\delta$

$\alpha = \tan A \quad \beta = \tan B \quad \gamma = \tan C$

$\delta = \tan D$.

$S_4 = \alpha\beta\gamma\delta$.

\# $S_3 = \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$.

$$\tan(A+B+C+D+E) = \frac{S_1 - S_3 + S_5}{1 - S_2 + S_4}$$

$tA\ tB\ tC\ tD\ tE$

$\alpha = \tan A \quad \beta = \tan B \quad \gamma = \tan C$

$\delta = \tan D$. $t = \tan E$.

$\alpha\beta\gamma\delta t$

$S_4 = \alpha\beta\gamma\delta + \alpha\beta\gamma t + \alpha\beta\delta t + \alpha\gamma\delta t + \beta\gamma\delta t$

$S_5 = \alpha\beta\gamma\delta t$

ex:

$x^3 - 7x^2 - 5x + 13 = 0$ — roots: $\tan A$, $\tan B$, $\tan C$.

** Quad. ka paap!

$S_1 = \tan A + \tan B + \tan C = \frac{7}{1} = 7$

$S_2 = \tan A \tan B + \tan B \tan C + \tan A \tan C = -5$

$S_3 = \tan A \tan B \tan C = -13$

$$\tan(A+B+C) = \frac{S_1 - S_3}{1 - S_2}$$

$$= \frac{7-(-13)}{1-(-5)} = \frac{20}{6} = \frac{10}{3}$$ ✓

Question

[Ans. : C, D]

If $\prod_{k=1}^{45}(1+\tan k^\circ) = 2^t$, then '$t$' is/are -



A Composite number

B Even number

C Prime number

D Odd number

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

add

$$\sin\underbrace{(A+B)}_{C} + \sin\underbrace{(A-B)}_{D} = 2\sin A \cos B$$

$\sin C + \sin D$ = add^{n} of

$2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$ = multip. of

$$A+B = C$$
$$A-B = D$$

add / sub.

$$2A = C+D \qquad 2B = C-D$$

$$A = \frac{C+D}{2} \qquad B = \frac{C-D}{2}$$

TRANSFORMATION FORMULAS

Transforming Sum / Difference into Product:

$$\sin C + \sin D = 2 \sin\frac{C+D}{2} \cos\frac{C-D}{2}$$

$$\sin C - \sin D = 2 \sin\frac{C-D}{2} \cos\frac{C+D}{2}$$

$$\cos C + \cos D = 2 \cos\frac{C+D}{2} \cos\frac{C-D}{2}$$

$$\cos C - \cos D = \ominus 2 \sin\frac{C+D}{2} \sin\frac{C-D}{2}$$



RAR

RAPID FIRE ROUND

(i) $\sin 2\theta - \sin 5\theta = 2\sin\left(\frac{2\theta-5\theta}{2}\right)\cos\left(\frac{2\theta+5\theta}{2}\right) = -2\sin\frac{3\theta}{2}\cos\frac{7\theta}{2}$ ($\frac{-3\theta}{2}$)

(ii) $\cos 8\theta + \cos 4\theta = 2\cos\left(\frac{8\theta+4\theta}{2}\right)\cos\left(\frac{8\theta-4\theta}{2}\right) = 2\cos 6\theta \cdot \cos 2\theta$

(iii) $\cos\theta - \cos 7\theta = -2\sin\left(\frac{\theta+7\theta}{2}\right)\sin\left(\frac{\theta-7\theta}{2}\right) = +2\sin 4\theta \sin(+3\theta)$

RAPID FIRE ROUND

$\cos(180^\circ - 5^\circ) = -\cos 5^\circ$

Basic Use:-

(i) Value of : $\cos 55^\circ + \cos 65^\circ + \cos 175^\circ$ = $2\cos 60^\circ \cos(-5^\circ) + (-\cos 5^\circ)$

$2 \times \frac{1}{2} \cos 5^\circ - \cos 5^\circ = 0.$

(ii) If $\sin 112^\circ + \cos 82^\circ = k \cos 52^\circ$ then $k = \sqrt{3}$.

$\cos 82^\circ = \sin(90^\circ - 82^\circ)$

$\sin 112^\circ + \sin 8^\circ = 2\sin\left(\frac{120^\circ}{2}\right) \times \cos\left(\frac{112^\circ - 8^\circ}{2}\right)$

$= 2 \times \frac{\sqrt{3}}{2} \cos 52^\circ$

$= \sqrt{3} \cos 52^\circ$


S.A.S.
Trigo. Terms in ⊕ OR ⊖
Convert in multiplication

Prove the following: $\frac{\sin A - \sin 5A + \sin 9A - \sin 13A}{\cos A - \cos 5A - \cos 9A + \cos 13A} = \cot 4A$

$$\text{L.H.S.} = \underbrace{\sin A - \sin 5A}_{C,D} + \underbrace{\sin 9A - \sin 13A}$$

$$2\sin\left(\frac{A-5A}{2}\right)\cos\left(\frac{A+5A}{2}\right) + 2\sin\left(\frac{9A-13A}{2}\right)\cos\left(\frac{9A+13A}{2}\right)$$

$$2\sin(-2A)\cos 3A + 2\sin(-2A)\cos 11A$$

$$-2\sin(2A)\{\cos 3A + \cos 11A\} = -2\sin 2A\,(2\cos 7A\cos 4A)$$

Prove the following: $\frac{\sin A - \sin 5A + \sin 9A - \sin 13A}{\cos A - \cos 5A - \cos 9A + \cos 13A} = \cot 4A$

$D^r = (\cos A - \cos 5A) - (\cos 9A - \cos 13A)$

$= +2\sin(+2A)\sin 3A - (+2\sin(+2A)\sin 11A)$

$2\sin 2A \{\sin 3A - \sin 11A\}$

$D^r = 2\sin 2A \; 2\sin(-4A)\cos 7A$

$\frac{N^r}{D^r} = \frac{+4\sin 2A \cos 7A \cos 4A}{+4\sin 2A \sin 4A \cos 7A}$

$= \cot 4A$

Hence Proved.

Question

If α in first quadrant = $\frac{\pi}{19}$. Find value of $\frac{\sin 23\alpha - \sin 3\alpha}{\sin 16\alpha + \sin 4\alpha}$.



TRANSFORMATION FORMULAS

Transforming Product into Sum / Difference:

- $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
- $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
- $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
- $2\sin A\sin B = \cos(A-B) - \cos(A+B)$



RAPID FIRE ROUND

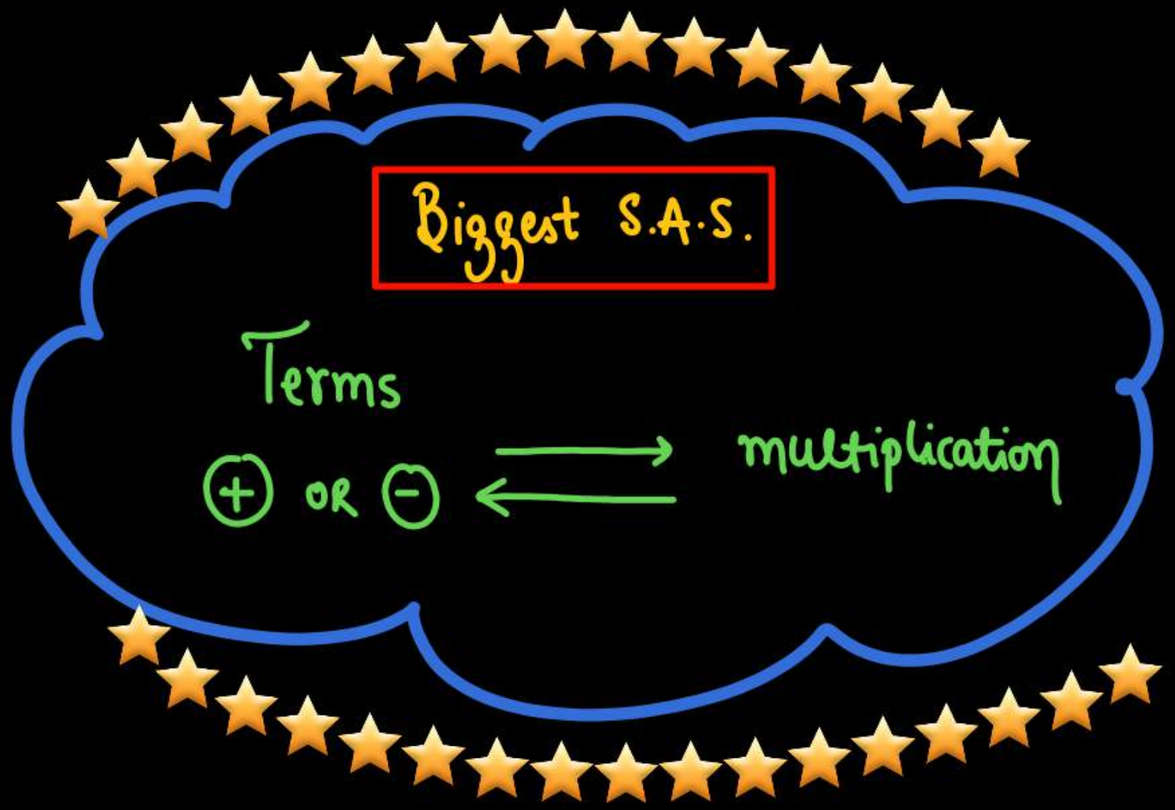
Formula laga key dikhao:

(i) $\frac{1}{2}(2\sin\overset{A}{3\theta}\cdot\cos\overset{B}{4\theta}) = \frac{1}{2}\left(\sin(3\theta+4\theta)+\sin(3\theta-4\theta)\right) = \frac{1}{2}\left(\sin 7\theta - \sin\theta\right)$ ✓

(ii) $2\cos\overset{A}{\frown 2\theta}\cdot\cos\overset{B}{\frown 6\theta} = \cos(8\theta)+\cos(4\theta)$ ✓

(iii) $2\sin\overset{A}{\frown\theta}\cdot\sin\overset{B}{\frown 3\theta} = \cos(\theta-3\theta)-\cos(\theta+3\theta)$

$= \cos 2\theta - \cos 4\theta$ ✓


Biggest S.A.S.
Terms
⊕ or ⊖
multiplication

Question

Prove the following: $\sin\frac{\theta}{2}\sin\frac{7\theta}{2}+\sin\frac{3\theta}{2}\sin\frac{11\theta}{2}=\sin2\theta\sin5\theta$

$$LHS=\frac{1}{2}\left(2\sin\underset{A}{\frac{\theta}{2}}\sin\underset{B}{\frac{7\theta}{2}}\right)+\frac{1}{2}\left(2\sin\underset{A}{\frac{3\theta}{2}}\sin\underset{B}{\frac{11\theta}{2}}\right)$$

$$\frac{1}{2}\left(\cos\left(\frac{\theta}{2}-\frac{7\theta}{2}\right)-\cos\left(\frac{\theta}{2}+\frac{7\theta}{2}\right)\right)+\frac{1}{2}\left(\cos\left(\frac{3\theta}{2}-\frac{11\theta}{2}\right)-\cos\left(\frac{3\theta}{2}+\frac{11\theta}{2}\right)\right)$$

$$\frac{1}{2}(\cos3\theta-\cos4\theta)+\frac{1}{2}(\cos4\theta-\cos7\theta)=\frac{1}{2}(\cos3\theta-\cancel{\cos4\theta}+\cancel{\cos4\theta}-\cos7\theta)$$

$$= \frac{1}{2}\left(\overset{C}{\cos 30} - \overset{D}{\cos 70}\right)$$

$$= \frac{1}{2}\left(+2\sin 50 \sin\frac{+40}{2}\right)$$

$$= \sin 50 \sin 20$$

HHP.

Question

Find the value of expression $2\cos\frac{\pi}{13}\cos\frac{9\pi}{13}+\cos\frac{3\pi}{13}+\cos\frac{5\pi}{13}$.

$\# \quad \frac{\pi}{13}=\theta$

$\pi = 13\theta$

$2\cos\theta\cdot\cos 9\theta + \overset{C}{\cos 3\theta} + \overset{D}{\cos 5\theta}$

$90^\circ = \frac{\pi}{2} = \frac{13\theta}{2}$

$\cos 90^\circ = 0$

$2\cos\theta\cdot\cos 9\theta + 2\cos 4\theta\cos\theta$

$2\cos\theta\left\{\overset{C}{\cos 9\theta} + \overset{D}{\cos 4\theta}\right\}$

$= 2\cos\theta\; 2\cos\frac{13\theta}{2}\cos\frac{5\theta}{2} = 0$

Question

Prove that $\frac{\sin A + 2\sin 3A + \sin 5A}{\sin 3A + 2\sin 5A + \sin 7A} = \frac{\sin 3A}{\sin 5A}$.



Question

Prove that $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{8}$

$$\text{LHS} = \frac{1}{2}\left(2\sin 20^\circ \sin 40^\circ\right)\sin 80^\circ$$

$$= \frac{1}{2}\left(\cos(20^\circ) - \cos 60^\circ\right)\sin 80^\circ$$

$$= \frac{1}{2}\left(\frac{1}{2}\left(2\cos 20^\circ \sin 80^\circ\right) - \frac{1}{2}\sin 80^\circ\right)$$

$$= \frac{1}{2}\left(\frac{1}{2}\left(\sin 100^\circ + \sin(60^\circ)\right) - \frac{1}{2}\sin 80^\circ\right)$$

$$= \frac{1}{2}\left(\frac{1}{2}\sin 100^\circ + \frac{1}{2}\sin 60^\circ - \frac{1}{2}\sin 80^\circ\right)$$

$$= \frac{1}{4} \times \frac{\sqrt{3}}{2}$$ HHPP.

$$\# \sin(100^\circ) = \sin(180^\circ - 80^\circ)$$

$$= \sin 80^\circ$$



Find value of : $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ $= 4$

$$\tan 9^\circ + \tan 81^\circ - (\tan 27^\circ + \tan 63^\circ)$$

$$\left(\frac{\sin 9^\circ}{\cos 9^\circ} + \frac{\sin 81^\circ}{\cos 81^\circ}\right) - \left(\frac{\sin 27^\circ}{\cos 27^\circ} + \frac{\sin 63^\circ}{\cos 63^\circ}\right)$$

$\sin 90^\circ = 1$

$$\sin(9^\circ + 81^\circ) \leftarrow \left(\frac{\sin 9^\circ \cos 81^\circ + \cos 9^\circ \sin 81^\circ}{\cos 9^\circ \cos 81^\circ}\right) - \left(\frac{\sin(27^\circ + 63^\circ)}{\cos 27^\circ \cos 63^\circ}\right)$$

$$\frac{1}{\cos 9^\circ \cos 81^\circ} - \frac{1}{\cos 27^\circ \cos 63^\circ}$$

$$= \frac{1}{\cos 9^\circ \cos 81^\circ} - \frac{1}{\cos 27^\circ \cos 63^\circ}$$

$$\frac{1}{\frac{1}{2}\cos 72^\circ} - \frac{1}{\frac{1}{2}\cos 36^\circ}$$

$$\frac{2}{\cos 72^\circ} - \frac{2}{\cos 36^\circ}$$

$$2\left(\frac{\cos 36^\circ - \cos 72^\circ}{\cos 72^\circ \cos 36^\circ}\right) = 2\left(\frac{2\sin 54^\circ \sin 18^\circ}{\cos 72^\circ \cos 36^\circ}\right)$$

$$= 4 \cdot \frac{\sin 54^\circ}{\cos 36^\circ} \cdot \frac{\sin 18^\circ}{\cos 72^\circ}$$

$$= 4 \cdot 1 \cdot 1 = 4$$

$\sin 54^\circ = \cos 36^\circ$

$\sin(90^\circ - 36^\circ)$

$\sin 18^\circ = \cos 72^\circ$

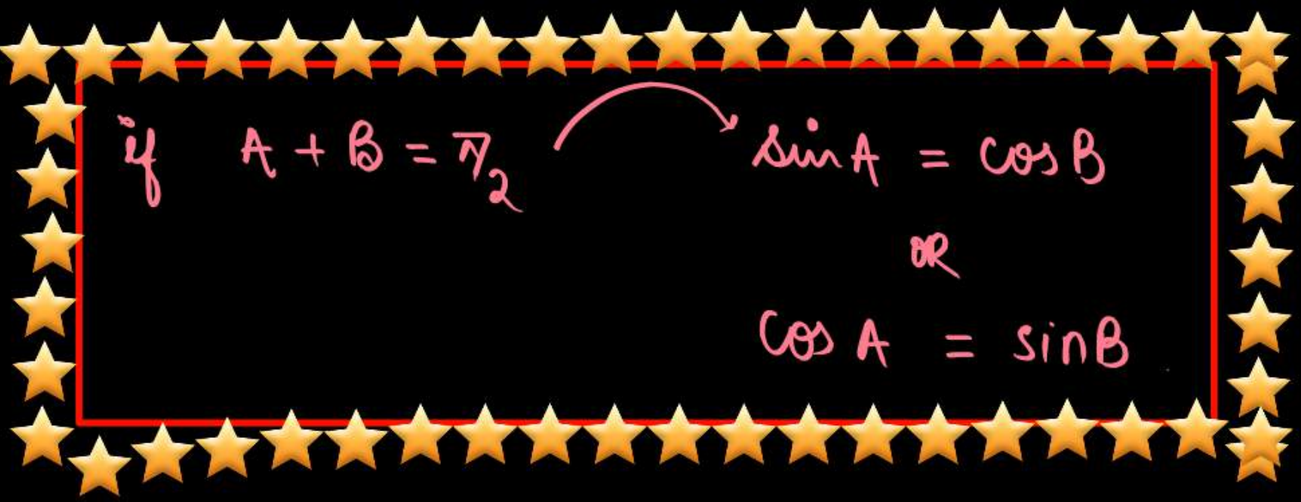
$$\# \frac{1}{2}\left(2\cos 9^\circ \cos 81^\circ\right) = \frac{1}{2}\left(\cos(9^\circ + 81^\circ) + \cos(9^\circ - 81^\circ)\right)$$

$$= \frac{1}{2}\left(0 + \cos 72^\circ\right)$$

$$= \frac{1}{2}\cos 72^\circ$$

$$\frac{1}{2}\left(2\cos 27^\circ \cos 63^\circ\right) = \frac{1}{2}\left(\cos 90^\circ + \cos 36^\circ\right)$$

$$= \frac{1}{2}\cos 36^\circ$$


if A + B = π/2
sin A = cos B
OR
cos A = sin B

Question [JEE-1984]

$$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$$



A $\frac{1}{2}$

B $\cos\frac{\pi}{8}$

C $\frac{1}{8}$

D $\frac{1+\sqrt{2}}{2\sqrt{2}}$

DO IT BY YOURSELF (DIBY)

$\frac{7\pi}{18} + \frac{\pi}{18} = \frac{8\pi}{18} = \frac{4\pi}{9}$ $\qquad$ $\frac{\pi}{2} - \frac{\pi}{9} = \frac{7\pi}{18}$

DIBY-18

$\cos^2\frac{7\pi}{18}$ $\qquad$ 1

$\sin^2\frac{\pi}{18} + \sin^2\frac{\pi}{9} + \sin^2\frac{7\pi}{18} + \sin^2\frac{4\pi}{9}$ equal

(A) 0 $\qquad$ (B) 1 $\qquad$ (C) −1 $\qquad$ (D) 2

$\cos^2\left(\frac{\pi}{2} - \frac{\pi}{9}\right) = \cos^2\frac{7\pi}{18}$

[Ans. : (D)]

DIBY-19

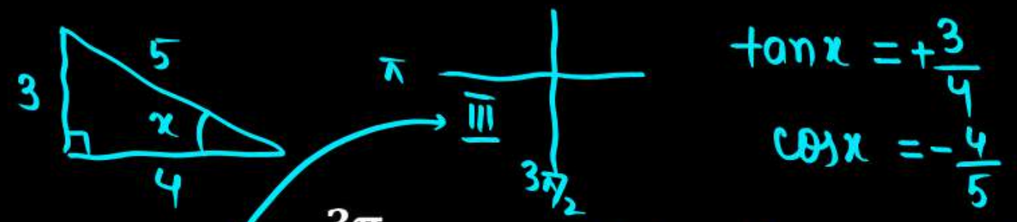
Prove that

1. $\tan 720° - \cos 270° - \sin 150° \cos 120° = \frac{1}{4}$

2. $\sin 780° \sin 480° + \cos 120° \sin 150° = \frac{1}{2}$

DIBY-20

Prove that $\sec\left(\frac{3\pi}{2} - \theta\right)\sec\left(\theta - \frac{5\pi}{2}\right) + \tan\left(\frac{5\pi}{2} + \theta\right)\tan\left(\theta - \frac{3\pi}{2}\right) = -1.$



DIBY-21

If $\sin x = -\frac{3}{5}$ where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to

(A) 109 (B) 108 (C) 19 (D) 18

$80\left(\frac{9}{16} - \left(\frac{-4}{5}\right)\right)$

$80\left(\frac{45+64}{80}\right) = 109$

[Ans. : (A)]

DIBY-22

1. $\sin(405^\circ) = \sin(4\times 90^\circ + 45^\circ) = +\sin(45^\circ) = \frac{1}{\sqrt{2}}$ [Ans. : $\frac{1}{\sqrt{2}}$]
2. $\sec(-1470^\circ)$ [Ans. : $\frac{2}{\sqrt{3}}$]
3. $\tan(-300^\circ)$ [Ans. : $\sqrt{3}$]
4. $\cot(585^\circ)$ [Ans. : 1]
5. $\operatorname{cosec}(-750^\circ)$ [Ans. : -2]
6. $\cos(-2220^\circ) = \cos(2220^\circ) = \cos(\overbrace{24\times 90^\circ}^{12\times 180^\circ} + 60^\circ) = +\cos 60^\circ = \frac{1}{2}$ [Ans. : 1/2]

DIBY-23

Prove that

1. $\frac{\sin 7\theta-\sin 5\theta}{\cos 7\theta+\cos 5\theta}=\tan\theta$

2. $\frac{\cos 6\theta-\cos 4\theta}{\sin 6\theta+\sin 4\theta}=-\tan\theta$

3. $\frac{\sin A+\sin 3A}{\cos A+\cos 3A}=\tan 2A$

4. $\frac{\sin 7A-\sin A}{\sin 8A-\sin 2A}=\cos 4A\sec 5A$

5. $\frac{\cos 2B+\cos 2A}{\cos 2B-\cos 2A}=\cot(A+B)\cot(A-B)$

6. $\frac{\sin 2A+\sin 2B}{\sin 2A-\sin 2B}=\frac{\tan(A+B)}{\tan(A-B)}$

7. $\frac{\sin A+\sin 2A}{\cos A-\cos 2A}=\cot\frac{A}{2}$

8. $\frac{\sin 5A-\sin 3A}{\cos 3A+\cos 5A}=\tan A$

9. $\frac{\cos 2B-\cos 2A}{\sin 2B+\sin 2A}=\tan(A-B)$

10. $\cos(A+B)+\sin(A-B)=2\sin(45^\circ+A)\cos(45^\circ+B)$

11. $\frac{\cos 3A-\cos A}{\sin 3A-\sin A}+\frac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}=\frac{\sin A}{\cos 2A\cos 3A}$



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians